CSE 428 · COMPUTER VISION PROJECT

Pet Segmentation & *Breed Classification*

A multi-task deep learning study on the Oxford-IIIT Pet dataset

TEAM

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HARDWARE

AMD RYZEN 5800xt
AMD Radeon RX 9060 XT · 16 GB VRAM
DDR4 RAM 16 GB 3200 MHz

PyTorch · ROCm

OVERVIEW

Problem & Goal

The challenge

Given an image of a cat or a dog, the model must:

- Predict a foreground/background mask separating the pet from the scene (binary segmentation).
- Predict the breed out of 37 fine-grained classes (12 dog + 25 cat breeds).

7,349 Total images	**37** Breed classes
256×256 Input size	**30** Training epochs

Two heads, one network: the segmentation decoder produces masks; a classification head off the U-Net bottleneck predicts the breed — trained jointly with a combined loss (Segmentation + 0.5·Classification).

Dataset — Oxford-IIIT Pet

Random samples with ground-truth masks

At a glance

37 breeds

25 cat + 12 dog breeds, ~200 images each.

Trimap masks

Pixel labels {1, 2, 3}; merged to a binary {pet, background} mask using {1, 3} → pet.

Splits

70% train / 15% val / 15% test, stratified by breed (seed = 42).

Augmentation pipeline

Horizontal flip, brightness/contrast jitter, small affine (±15°, ±5% translate, 0.9–1.1× scale), Gaussian noise.

METHOD

Architecture — Multi-task U-Net

A standard encoder/decoder U-Net with a classifier head branching off the bottleneck. The Attention U-Net adds attention gates on every skip connection.

U-Net

Custom · joint segmentation + classification

- Encoder: 5 stages (64→1024 ch)
- Decoder: transpose-conv up-sampling with concat skip-connections
- Seg head: 1×1 conv → mask logits
- Cls head: bottleneck → MLP → 37 logits
- Combined loss: BCE(seg) + 0.5·CE(cls)

Attention U-Net

Custom · gated skips

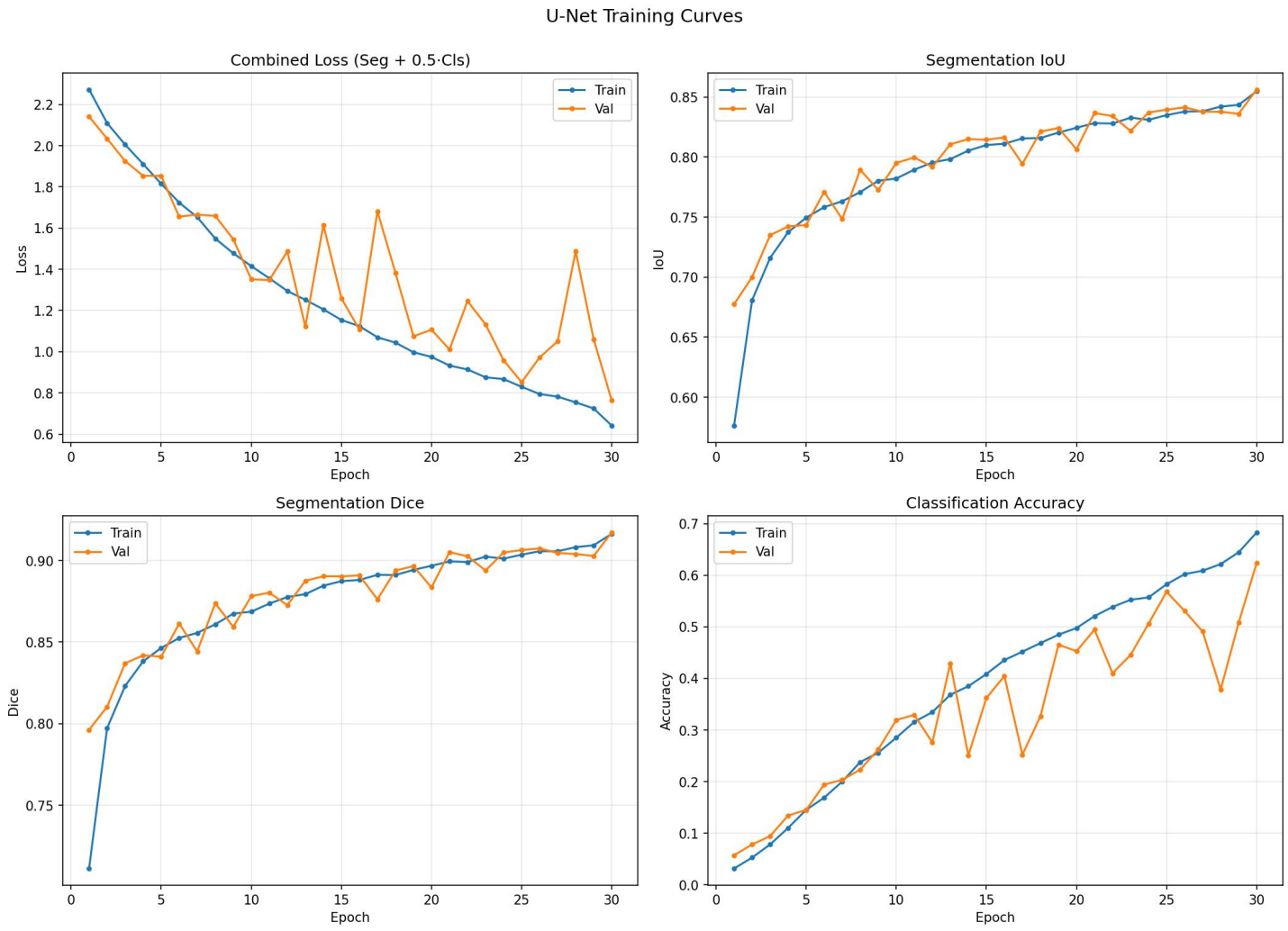
- Same encoder/decoder layout
- Attention gates re-weight encoder features before concatenation
- Sharpens object boundaries, suppresses background noise
- Same dual-head (seg + cls) and combined loss

Pretrained classifiers

ImageNet · classification only

- MobileNetV3-Large
- EfficientNet-B0
- DenseNet-121
- Final layer swapped for 37-way head; remainder fine-tuned
- Used as bonus comparison for the classification task only

U-Net — Training curves



Takeaways

- Segmentation IoU climbs steadily to ~0.86 with no signs of overfitting.
- Dice plateaus near 0.92 — boundaries are sharp.
- Classification accuracy is the bottleneck: ~0.62 by epoch 30, with visible val-set noise (single bottleneck → 37 fine-grained breeds).
- Loss curves still trending down → could benefit from more epochs or a stronger classifier head.

U-Net — Sample predictions

Original class - "yorkshire terrier (Dog)"
Predicted class - "yorkshire terrier (Dog)"
IoU - 87.5%

Original image

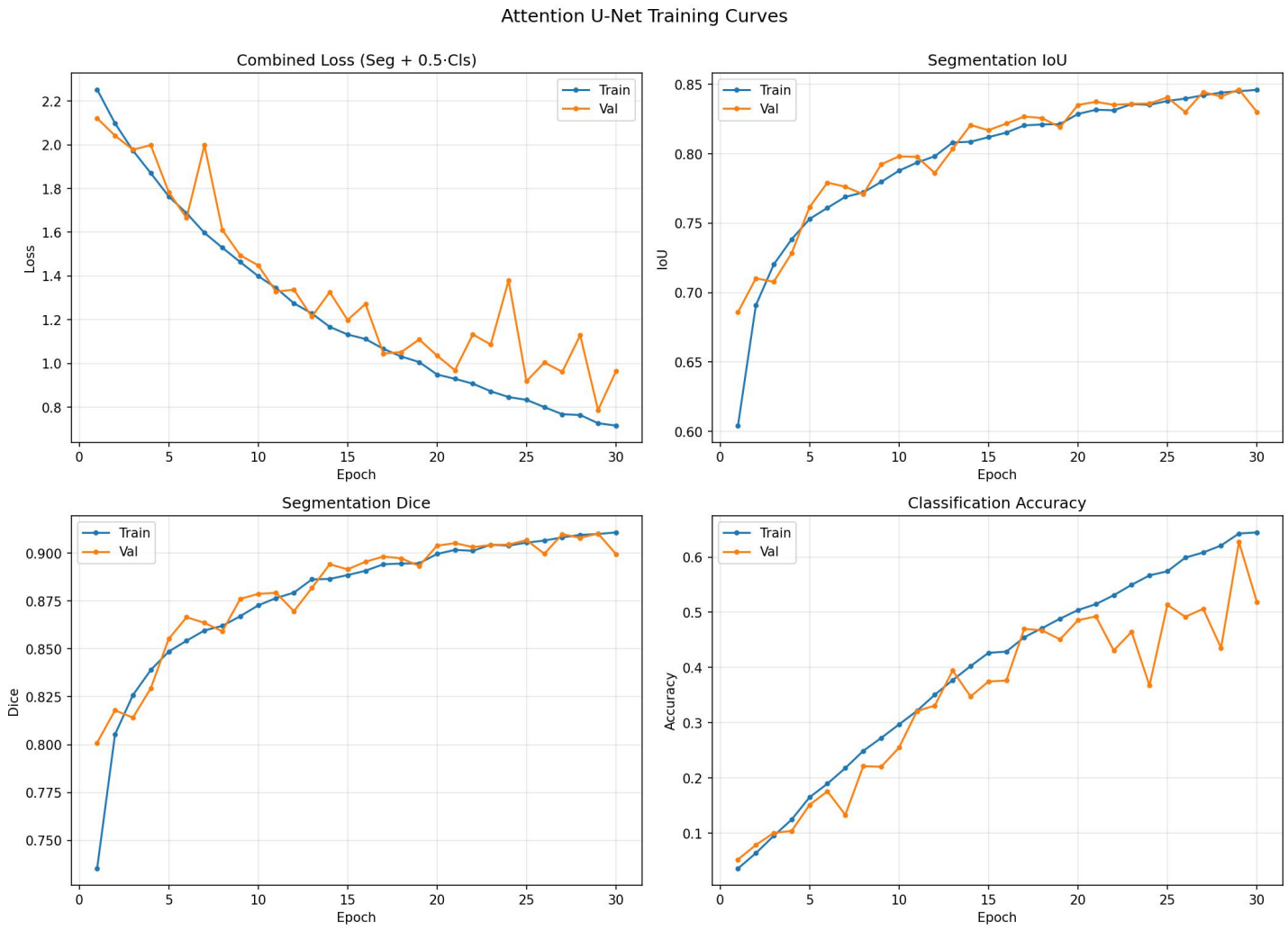
Image + True mask overlay

Image + Predicted mask overlay

What this shows

Yorkshire terrier — correctly classified, IoU 87.5%. Even with the pet in the frame on a unusual angle and a busy background (yellow object, tile floor), the model isolates the animal and produces a clean dual-region mask skipping the red cloth.

Attention U-Net — Training curves



Takeaways

- Slightly smoother loss curve early on — gates stabilize gradient flow through skips.
- Final mIoU and Dice land in the same range as the base U-Net; no clear segmentation win on this dataset.
- Classification accuracy similarly tops out around 0.61 — same single-bottleneck bottleneck as base.
- Visual inspection shows tighter mask boundaries on a few hard cases (fur edges, occlusion).

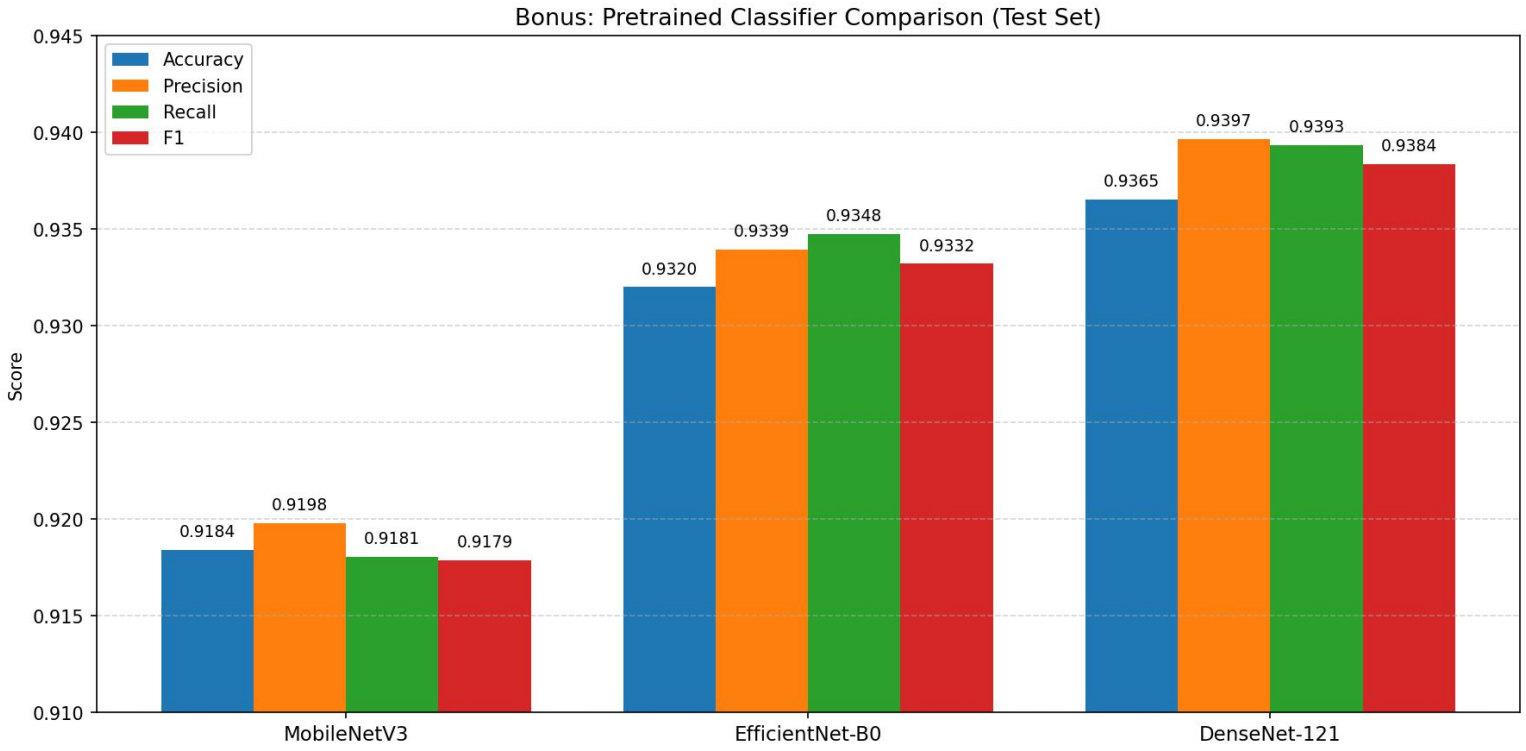
Attention U-Net — Sample predictions

Original class - "Abyssinian (Cat)"
Predicted class - "British Shorthair (Cat)"
IoU - 77.1%

An interesting failure case

Abyssinian misclassified as British Shorthair — mask IoU 77.1%. Both breeds are short-haired grey cats, so the breed mix-up is plausible. This case the small predicted-mask leaks on the right side and left side; the attention gates haven't fully suppressed background noise here.

Bonus — Pretrained classifier comparison



Three well-established classifiers

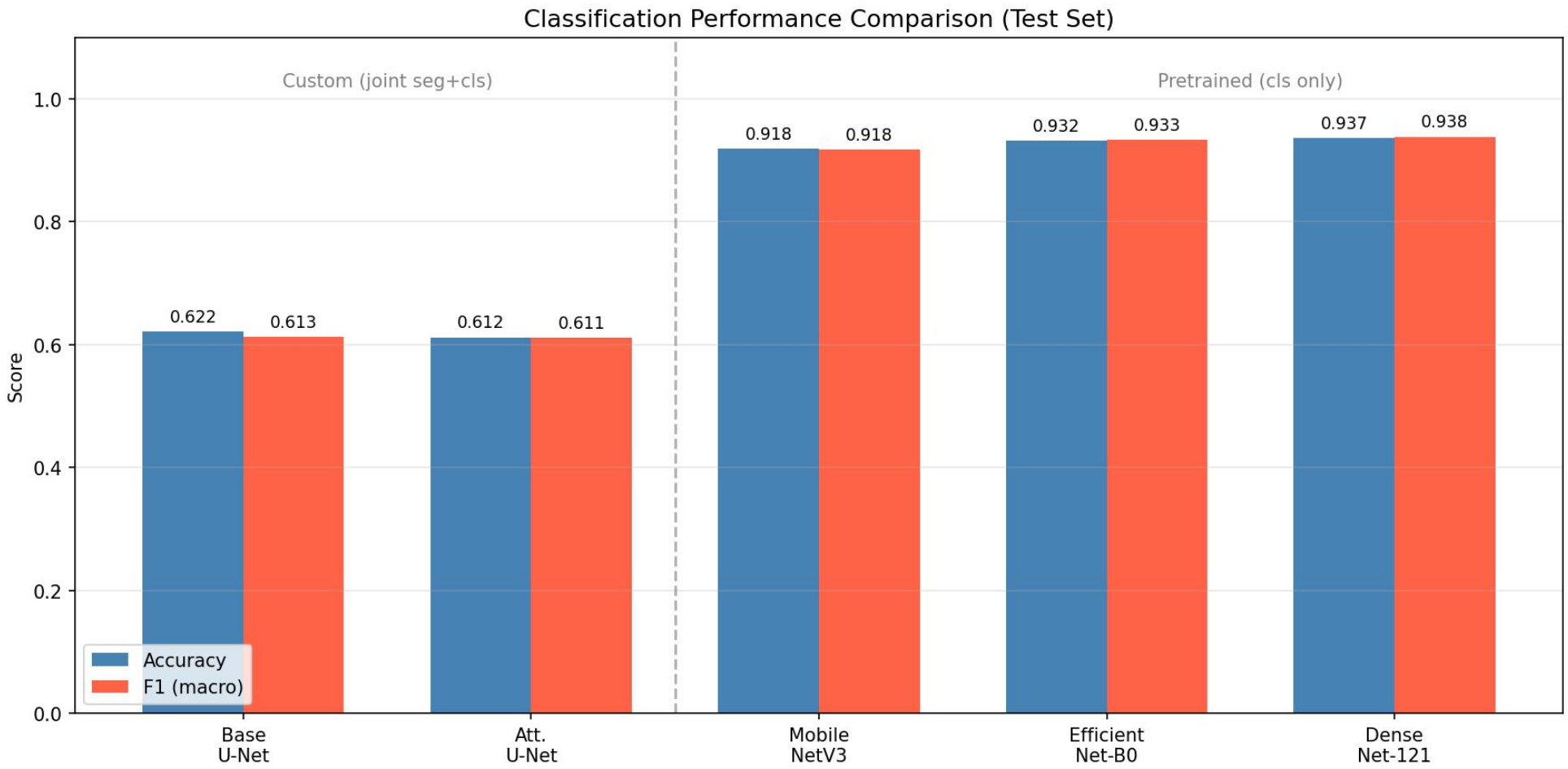
MobileNetV3-Large
Acc 0.918 · F1 0.918 — fastest, smallest.

EfficientNet-B0
Acc 0.932 · F1 0.933 — best balance.

DenseNet-121
Acc 0.937 · F1 0.938 — top performer.

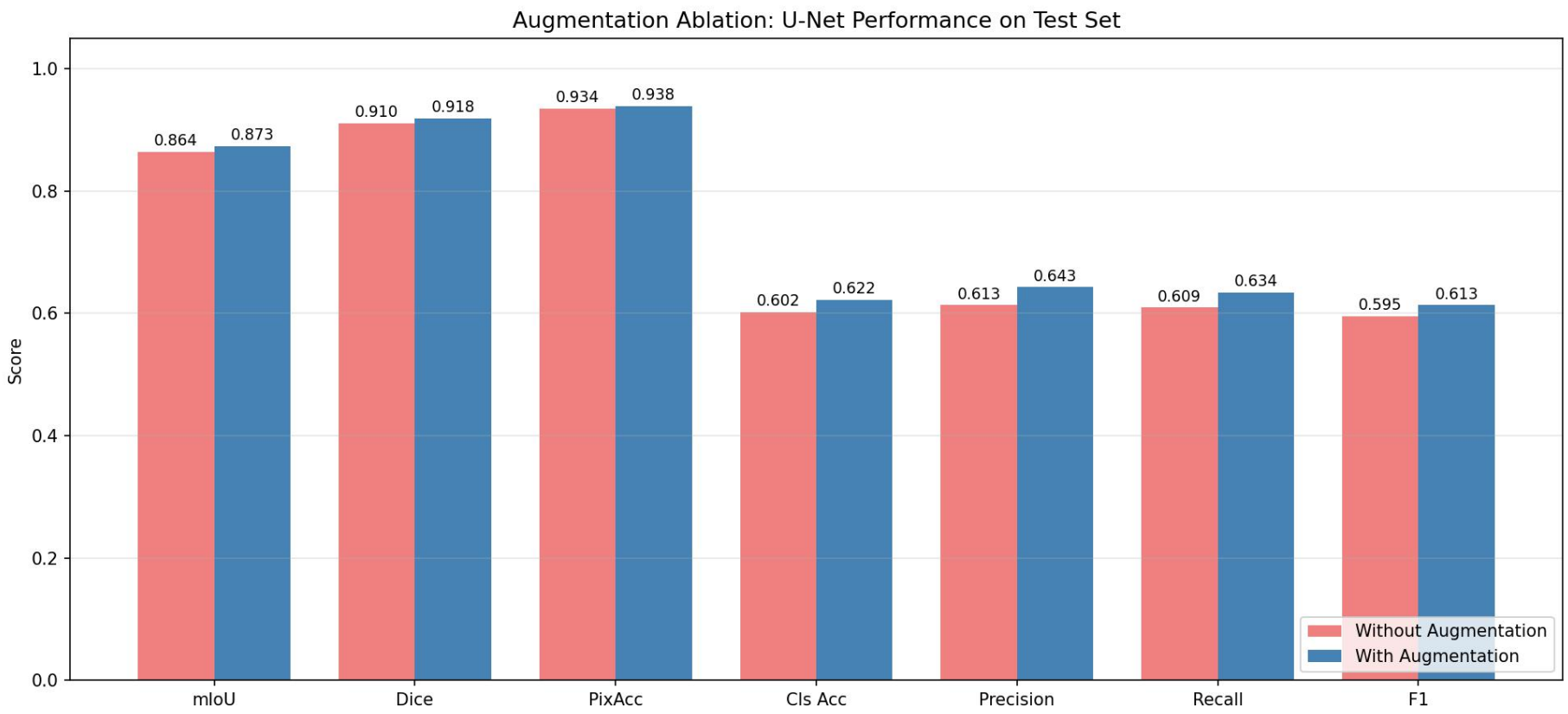
Headline result — pretrained classifiers (~93%) outperform our custom multi-task heads (~62%) by a wide margin on classification, suggesting the U-Net bottleneck is a weak feature extractor for fine-grained breed Identification.

Custom vs pretrained — full comparison



Pretrained ImageNet weights provide a feature space that's already separable for fine-grained breeds — the gap to our jointly-trained U-Net heads is consistent across all metrics.

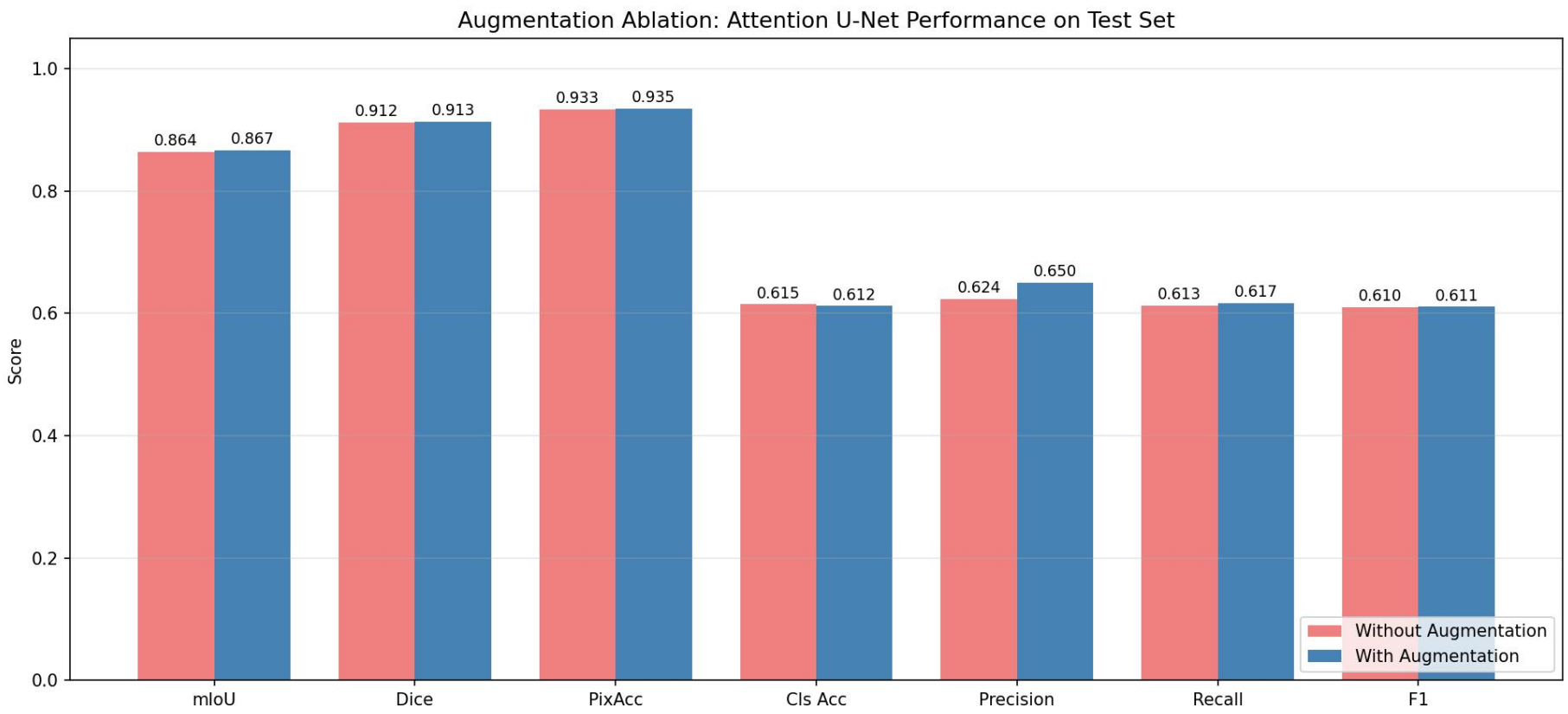
Augmentation ablation — U-Net



Effect of augmentation

- Segmentation: small but consistent gain — mIoU +0.009, Dice +0.008.
- Classification gains more in relative terms — F1 +0.018, Cls Acc +0.020.
- Worth the extra cost: aug pipeline adds < 5% per-epoch time.

Augmentation ablation — Attention U-Net



Effect of augmentation

- Segmentation barely moves — Attention gates already provide a regularising effect.
- Classification metrics flat or marginally noisier (Cls Acc −0.003, F1 +0.001).
- Suggests augmentation and attention partially cover the same regularisation budget.

SUMMARY

Final results — test set

Model	mIoU	Dice	PixAcc	Cls Acc	F1 (macro)
U-Net (aug)	0.873	0.918	0.938	0.622	0.613
U-Net (no aug)	0.864	0.910	0.934	0.602	0.595
Attention U-Net (aug)	0.867	0.913	0.935	0.612	0.611
Attention U-Net (no aug)	0.864	0.912	0.933	0.615	0.610
MobileNetV3-Large	—	—	—	0.918	0.918
EfficientNet-B0	—	—	—	0.932	0.933
DenseNet-121	—	—	—	0.937	0.938

Cells marked "—" are not applicable: pretrained classifiers do not produce segmentation masks.

Best segmentation: **U-Net (aug) at 0.873 mIoU**

Best classification: **DenseNet-121 at 0.938 F1**

UNET & ATT UNET models trained for 30 epochs at 256×256 · batch size 32 · Learning rate 1e-4 · AMD Radeon RX 9060 XT
Other classifier heads are trained for 15 epochs at 256×256 · batch size 32 · Learning rate 1e-4 · AMD Radeon RX 9060 XT

CONCLUSIONS

What we learned

01

Segmentation works well

Both U-Net variants reach ~0.87 mIoU and ~0.92 Dice. Attention adds qualitative improvements on hard boundaries.

02

Classification needs a stronger head

The shared U-Net bottleneck caps fine-grained accuracy at ~0.62. Pretrained classifiers reach 0.92–0.94 on the same data.

03

Augmentation helps modestly

Strongest effect on plain U-Net (Cls Acc +2%, F1 +1.8%). Marginal on Attention U-Net — gates already regularise.

Thank you.

Code, weights and notebook available on request · CSE 428 Project